Samsung Research

Quantum-inspired AI

PRISM Worklet

| 06/01/2026 | SRI-B

PRISM – Quantum AI

Problem Statement	Quantum-inspired Annealing for Multi-stage Reasoning
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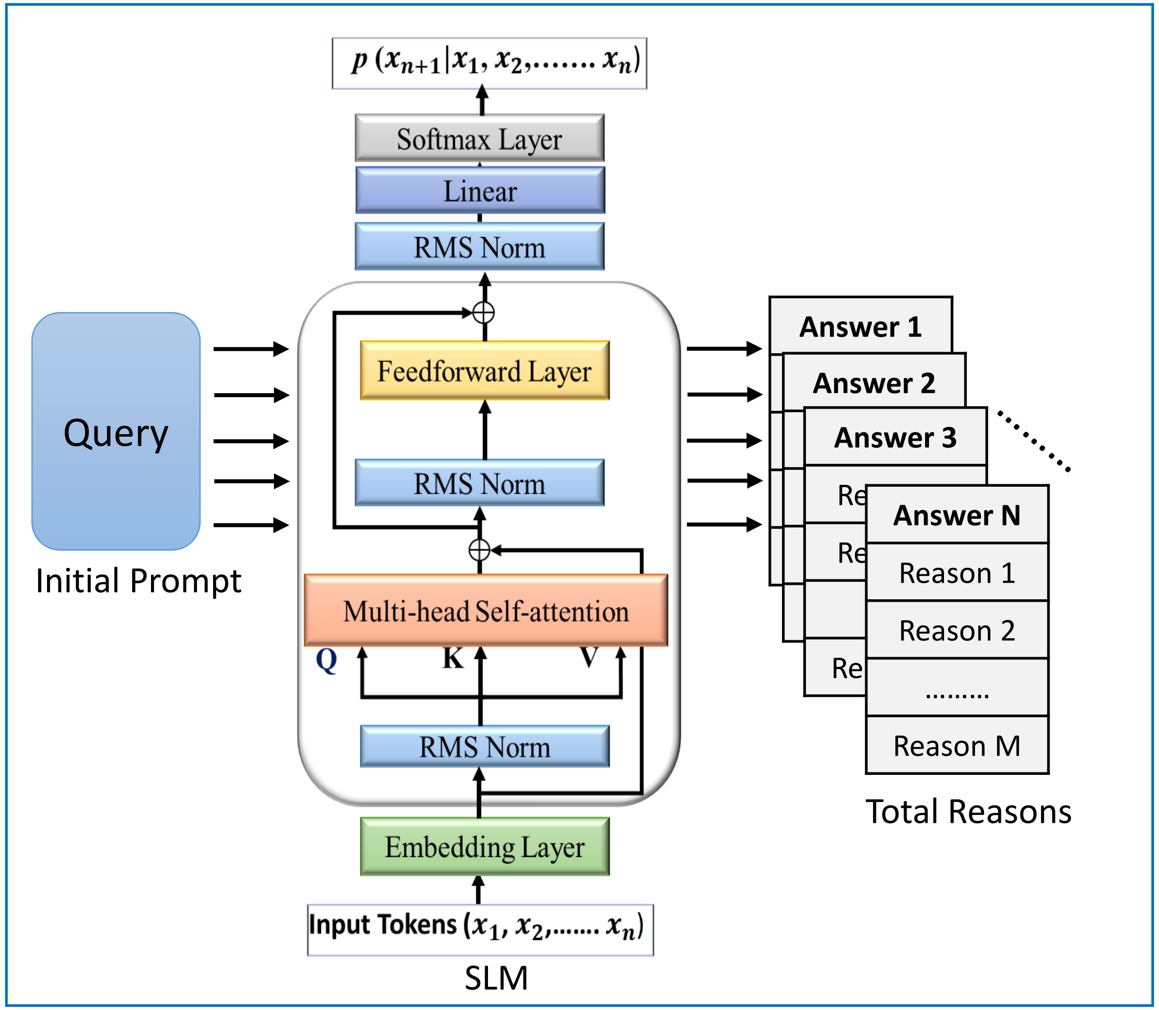
University	
Faculty	

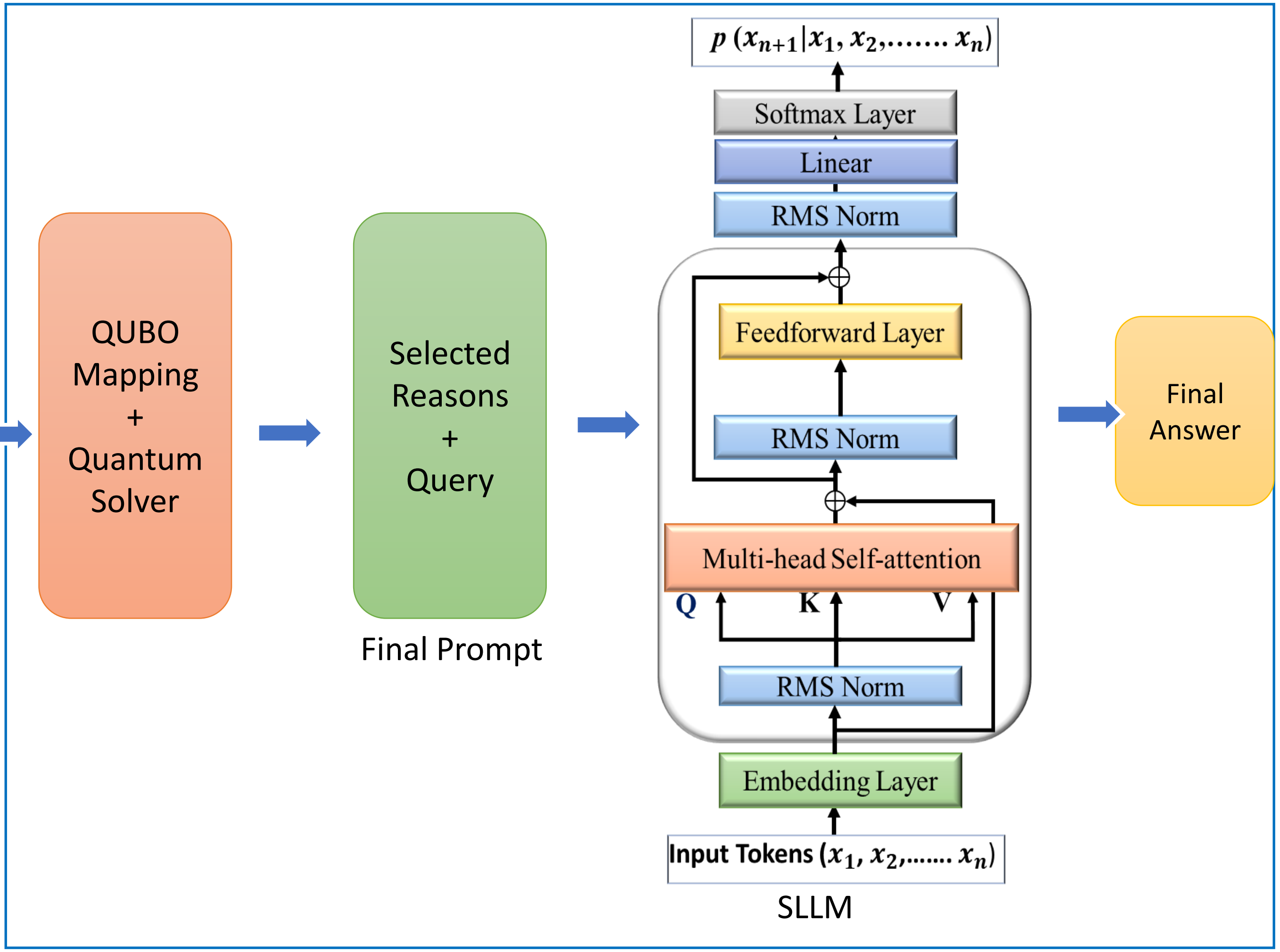
Project Summary

Samsung Research

Goal: Design and Implementation of Quantum-inspired Annealing for Multi-stage Reasoning using Small Language Models (SLMs)



Sampling Process



Selecting Reasons and Retrieving the Final Answers

Proposed Approach

1. Literature Review on Large Language Models and Reasoning [1][2][3][4]

2. Understanding the Recent variants of Chain of Thought (CoT) prompting (Self-Consistency [5], Tree-of-Thoughts (ToT)[6], and Voting-based Reasoning [7])

3. Understanding Quadratic Unconstrained Binary Optimization (QUBO) for combinatorial reasoning optimizations and implementations in D-wave Quantum Annealer [8, 9, 10]

4. Optimize a Multi-stage Reasoning using **Quantum-inspired Annealing** pipeline with SLMs (3B+) to achieve higher compression with minimal accuracy loss.

References

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6. S. Yao, D. Yu, J. Zhao, et al. "Tree of thoughts: deliberate problem solving with large language models" In Proc. of the 37th Int. Conf. on Neural Inf. Proc. Sys. (NIPS 2023). https://dl.acm.org/doi/abs/10.5555/3666122.3666639
7. W. Wang, Y. Wang, and H. Huang, "Ranked voting based self-consistency of large language models" In Find. of the Association for Computational Linguistics (ACL 2025). https://aclanthology.org/2025.findings-acl.744/
8. M. Esencan, et al. "Combinatorial reasoning: selecting reasons in generative AI pipelines via combinatorial optimization." arXiv preprint (2024). https://doi.org/10.48550/arXiv.2407.00071
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10. P. Chandarana, et al. "Runtime Quantum Advantage with Digital Quantum Optimization." arXiv preprint (2025). https://doi.org/10.48550/arXiv.2505.08663

Roadmap

Feb-May 2026

1. Literature Review on Large Language Models and Reasoning
2. Understanding the Recent variants of Chain of Thought (CoT) prompting algorithms

3. Understanding Quadratic Unconstrained Binary Optimization (QUBO) and present the analysis report

Understanding Reasoning Algorithms for Large Language Models

Jun-Jul 2026

1. Configure and Optimize the Multi-stage Reasoning using Quantum-inspired Annealing in SLM pipeline with the help of HPC (Nvidia A100)
2. Applying and integrating to SLMs

3. Explorations for Quantum-inspired Annealing Algorithms for optimization of the Hyper-parameters of Llama-3.2-3B with HPC
4. Identify and develop custom extensions of the model for simulating on CPU

Development, Optimization, and Quantum-inspired Annealing Algorithms explorations for Multi-stage Reasoning

Aug-Sep 2025

1. Performance gains on optimized module: >2x from Llama-3.2-3B+

Experimental study, developments and documentation of the explorations and results